SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
Student Name:	P. Venkata Kiran	**Reg. No.:**	192525425
Section / Batch:	2025	**Date:**	21/10/2025

Code of Conduct

*I **P. Venkata Kiran/192525425** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: *Venkata Kiran*

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L = 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

1. Histogram Equalization

Given pixel intensity values:

10, 10, 20, 20, 20, 30, 30, 40

Total number of pixels, N=8 and L=256.

Step 1: Construct the histogram

Intensity (\(r\))	**Frequency (\(f\))**	**Probability (\(f/N\))**	**CDF**
10	2	0.25	0.25
20	3	0.375	0.625
30	2	0.25	0.875
40	1	0.125	1.00

Step 2: Apply histogram equalization

Formula:

$$s = \text{round}\big((L-1) \times CDF\big)$$

Since \(L=256\),

$$s = \text{round}(255 \times CDF)$$

- For intensity 10:

$$s = \text{round}(255 \times 0.25) = 64$$

- For intensity 20:

$$s = \text{round}(255 \times 0.625) = 159$$

For intensity 30:

$$s = \text{round}(255 \times 0.875) = 223$$

- For intensity 40:

$$s = \text{round}(255 \times 1) = 255$$

Step 3: New intensity values

Original intensity	New intensity
10	64
20	159
30	223
40	255

The equalized pixel sequence is:

64, 64, 159, 159, 159, 223, 223, 255

Comparison: Before equalization, the intensity values lie within a narrow range of 10 to 40. After equalization, they spread across 64 to 255. This increases the intensity range and can improve image contrast.

2. Averaging Filter and Median Filter

Given image:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 100 & 20 \\ 10 & 20 & 30 \end{bmatrix}$$

Step 1: Apply the averaging filter

Formula:

$$\text{Average} = \frac{\text{Sum of all pixels}}{9}$$

Substituting the values:

$$= \frac{10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30}{9}$$

$$= \frac{260}{9} = 28.89$$

Therefore, the new centre pixel is approximately 29.

Replacing the centre pixel while keeping the other pixels unchanged:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 29 & 20 \\ 10 & 20 & 30 \end{bmatrix}$$

Step 2: Apply the median filter

Arrange all nine values in ascending order:

$$10,10,20,20,20,20,30,30,100$$

There are nine values, so the median is the fifth value.

$$\text{Median} = 20$$

The new centre pixel is 20.

Result:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 20 & 20 \\ 10 & 20 & 30 \end{bmatrix}$$

Step 3: Comparison

- Averaging filter: Calculates the mean of neighbouring pixels. It smooths the image but can blur edges and fine details.
- Median filter: Selects the middle value after sorting. It removes isolated extreme values while preserving edges better.

The median filter is better for this example because it reduces the effect of the unusually high centre value, 100, without allowing it to strongly influence the result.

3. Laplacian Filter for Image Sharpening

Given intensity matrix:

$$I = \begin{bmatrix} 50 & 50 & 50 \\ 50 & 100 & 50 \\ 50 & 50 & 50 \end{bmatrix}$$

Laplacian filter mask:

$$K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Step 1: Calculate the Laplacian response

Multiply corresponding pixels and add the results.

$$R = (0 \times 50) + (-1 \times 50) + (0 \times 50)$$
$$+(-1 \times 50) + (4 \times 100) + (-1 \times 50)$$
$$+(0 \times 50) + (-1 \times 50) + (0 \times 50)$$
$$R = 400 - 50 - 50 - 50 - 50$$
$$\boxed{R = 200}$$

Step 2: Sharpen the centre pixel

Formula:

$$\text{New intensity} = \text{Original intensity} + R$$

$$= 100 + 200$$

$$\boxed{\text{New centre intensity} = 300}$$

Since 300 exceeds the maximum 8-bit intensity of 255, it would be clipped to 255 if the result is stored directly in an 8-bit image.

Step 3: Explanation

The Laplacian filter detects rapid changes in intensity by comparing the centre pixel with its neighbours. It produces a strong response where intensity changes sharply. Adding this response to the original image enhances edges, boundaries, and fine details.

The Laplacian response is 200, and the calculated sharpened centre intensity is 300 before clipping.

4. Global Thresholding and Binary Image

Given intensity matrix:

$$I = \begin{bmatrix} 40 & 60 & 120 \\ 80 & 150 & 200 \\ 30 & 100 & 180 \end{bmatrix}$$

Threshold value:

$$T = 100$$

The rule is:

- If pixel intensity is greater than or equal to 100, assign 1.
- If pixel intensity is less than 100, assign 0.

Step 1: Apply thresholding

- First row: 40 → 0, 60 → 0, 120 → 1
- Second row: 80 → 0, 150 → 1, 200 → 1
- Third row: 30 → 0, 100 → 1, 180 → 1

Step 2: Construct the binary image

$$\boxed{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}}$$

Step 3: Count foreground and background pixels

Foreground pixels (value 1):

$$120,\ 150,\ 200,\ 100,\ 180$$

Number of foreground pixels = 5

Background pixels (value 0):

$$40,\ 60,\ 80,\ 30$$

Number of background pixels = 4

Total pixels:

$$5 + 4 = 9$$

Step 4: Explanation and limitation

Global thresholding separates an image into foreground and background using a single threshold. In this example, pixels with intensity 100 or greater become foreground pixels. One limitation is that it may perform poorly when illumination varies across the image or when foreground and background intensities overlap.

The binary image contains 5 foreground pixels and 4 background pixels.

5. Sobel Operator for Edge Detection

Given intensity matrix:

$$I = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 90 \\ 10 & 50 & 90 \end{bmatrix}$$

Sobel masks:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Step 1: Calculate the horizontal gradient \(G_x\)

Multiply corresponding elements and add:

$$G_x = (-10 + 10) + (-20 + 180) + (-10 + 90)$$

$$G_x = 0 + 160 + 80$$

$$\boxed{G_x = 240}$$

Step 2: Calculate the vertical gradient \(G_y\)

$$G_y = (-10 - 20 - 10) + (10 + 100 + 90)$$

$$G_y = -40 + 200$$

$$\boxed{G_y = 160}$$

Step 3: Calculate gradient magnitude

Formula:

$$G = \sqrt{G_x^2 + G_y^2}$$

Substituting the values:

$$G = \sqrt{240^2 + 160^2}$$

$$G = \sqrt{57600 + 25600}$$

$$G = \sqrt{83200}$$

$$\boxed{G \approx 288.44}$$

Step 4: Determine the dominant edge direction

The gradient angle is:

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{160}{240}\right)$$

$$\boxed{\theta \approx 33.69^\circ}$$

Since $G_x = 240$ is greater than $G_y = 160$, the intensity changes more strongly in the horizontal direction. Therefore, the detected edge is predominantly vertical. The gradient direction is approximately $33.69°$measured from the positive horizontal axis.

Step 5: Explanation

The Sobel operator calculates intensity gradients in two directions. A large gradient magnitude indicates a strong intensity change and a possible edge. These detected edges help identify object boundaries and separate different regions of an image during segmentation.

$G_x = 240$, $G_y = 160$, gradient magnitude $= 288.44$ and the dominant edge orientation is approximately vertical.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____